

SCIENTIFIC MANAGEMENT



Henry L Gantt

EVOLUTION OF MANAGEMENT THEORIES

I. Classical Management Theory (1880-1930)

1. Scientific Management by F.W Taylor.
2. Administrative Management by Henri Fayol.
3. Bureaucratic Management by Max Webber.

II. Neo Classical Management Theory (1930-1950)

1. Human Relations Movement
2. Behavioral Sciences Movement

III. Modern Management Theory (1950 Onwards)

1. Quantitative Approach
2. System Approach
3. Contingency Approach

CLASSICAL AMANAGEMENT THEORY

1. Scientific Management

- **Frederick W. Taylor**(Chief engineer in Bethlehem Steel Company, USA) – **father of scientific management.**
- According to Taylor, Scientific management is an art of knowing exactly what you want your men to do & seeing that they do it in the best & cheapest way

STEPS IN SCIENTIFIC MANAGEMENT

1. Analyze the work scientifically.
2. Divide work & responsibility.
3. Select workers best suited to perform specific tasks.
4. Provide guidelines for worker performance.
5. Achieve support & corporation from workers.

PRINCIPLES OF SM

1. Replacing rules of thumb with science.
2. Achieving cooperation of human beings.
3. Working for maximum output.
4. Developing all workers to the fullest extent possible for their own and their company's highest prosperity.

TECHNIQUES OF SM

1. Scientific study and planning of work.
2. Scientific selection, placement and training.
3. Standardization of raw materials, machines and working conditions.
4. Division of responsibility b/w managers and workers.
5. Wage incentives.

CRITICISM IN SM

1. Unrealistic assumption that worker is a economic man.
2. Workers are considered as a factor of production which is lifeless.
3. Close supervision and control causes work stress in employees.

MODERN MANAGEMENT THEORY

1. Quantitative approach
2. System approach
3. Contingency approach

CONTRIBUTIONS OF F W TAYLOR



- Taylor pioneered the **time & motion study**.
- Taylor advocated scientific study of the task – which led to the concept of **scientific management**.
- Another concept of Taylor was ‘**A Fair Days Task**’
- The concept of **functional organisation** was suggested by Taylor
- Taylor suggested a wage incentive scheme known as ‘**Taylor's Differential Piece Rate Plan**’.

CONTRIBUTIONS OF GANTT

- Gantt's contributions were more in the form of **refinements rather than fundamental concepts**
- He improved Taylor's 'Differential Piece Rate System' & brought out his '**Task and Bonus Plan**'
- Gantt developed the daily balance chart, now known as the **Gantt Chart**

- Gantt pleaded for **wider recognition of human factor** in management
- Gantt also pleaded for a **policy of preaching & teaching workman** rather than driving the workers like cattle
- Gantt was of the opinion that **emphasis should be placed on service rather than profits**



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CONTRIBUTIONS OF GILBERTH

- **Time study** -It is direct observation of a task, using a time keeping deice(stop watch, video camera etc.) to record the time taken to accomplish a task.
- **Motion/Method study** – It is the analysis of work motion, consisting in part of filming the details of a worker's activities and their body posture while recording the time.



FRANK GILBRETH

- Frank & Lillian Gilbreth did a lot of work in order to improve work method & to discover one best method of establishing it – their main field of interest was **movement/motion study**
- In 1917, Gilbreth suggested the 1st **definition of motion study** as “ the science of eliminating wastefulness resulting from unnecessary, ill directed & inefficient motions”
- Gilbreth developed ‘**threbligs**’ – a list of **18 basic motions** to help analyse any workers movement
- He developed **micro motion study & simo chart.**
- Gilbreth applied motion analysis to office procedures.
- Gilbreth & Lillian, based on their study on **fatigue & its elimination** found out that fatigue can be eliminated by **lighting the load, spacing the work & by introducing rest periods**

THERBLIGS

- It divides the human activities into micro-motions called **Therbligs**.

MICRO-MOTION STUDY

➤ *Therbligs*

 Search

 Find

 Select

 Grasp

 Hold

 Transport Loaded

 Transport Empty

 Position

 Assemble

 Use

 Disassemble

 Inspect

 Preposition

 Release Load

 Unavoidable Delay

 Avoidable Delay

 Plan

 Rest

NUT,BOLT AND WASHER ASSEMBLY

<i>Left Hand Description</i>	<i>Symbol (L.H.)</i>	<i>Symbol (R.H.)</i>	<i>Right Hand Description</i>
1. Move to bolt	⇒	⇒	Move to first washer.
2. Grasp the bolt	○	○	Pickup/grasp the washer.
3. Move to position	⇒	⇒	To position
4. Hold	▽	○	Assembled to bolt.
5.		⇒	Move to second washer.
6.		○	Pickup/grasp the washer.
7.		⇒	To position
8.		○	Assembled to bolt.
9.		⇒	Move to nut.
10.		○	Pick up the nut.
11. Move to box	⇒	⇒	To position.
12. Aside to box	○	○	Assembled to bolt



Classification of Therbligs

Effective therbligs:

- Transport empty
- Grasp
- Transport loaded
- Release load
- Use
- Assemble
- Disassemble
- Inspect
- Rest

Ineffective therbligs:

- Hold
- Pre-position
- Position
- Search
- Select
- Plan
- Unavoidable delay
- Avoidable delay